

ASE230 Project 1 - LMS API Tutorial

How to Use Our School Website API

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 = You need to be logged in

How to Log In



Getting Your Login Token

- **What it is:** A special password that lasts 24 hours
- **How to use it:** Put it in the header like this: `Authorization: Bearer <token>`
- **Who can use it:** Students, teachers, and admins

Example:

```
curl -H "Authorization: Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9..."
```

How to Sign Up

POST /api/users/register

What it does: Create a new account

Do you need to be logged in? No

What to send: application/json

What to send:

```
{
  "username": "johndoe",
  "email": "john@example.com",
  "password": "password123",
  "first_name": "John",
  "last_name": "Doe",
  "role": "student"
}
```

Sign Up - What You Get Back

If it worked (201 Created):

```
{
  "success": true,
  "message": "User registered successfully",
  "token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9...",
  "user": {
    "id": 5,
    "username": "johndoe",
    "email": "john@example.com",
    "first_name": "John",
    "last_name": "Doe",
    "role": "student"
  }
}
```

bigbrain9842-796981.postman.co/workspace/83f4eff3-7005-4483-8c90-bc077f57cb29/request/49376604-e3190f57-35f5-4191-9962-7c639892319b?tab=body

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My Collection

GET Get data

POST Post data

POST Get data

Save Share Send

Params Authorization Headers (11) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "username": "testuser12",
3   "email": "test12@example.com",
4   "password": "password123",
5   "first_name": "Test",
6   "last_name": "User"
7 }
```

Body Cookies (1) Headers (15) Test Results (0/1) 201 Created 1.08 s 966 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "User registered successfully",
4   "token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJ1c2VyX2lkIjoiaWIsInVzZXJyYW11IjoidGVzdHVkZW50IiwiaWF0IjoxNzYwMDA1NDY5LCJleHAiOjE3NjA4OTE4Nj19.hikXiaqtg8Hg2to6kPzVigxqegimQQ2hzUwaiQ2t5wA",
5   "user": {
6     "id": "6",
7     "username": "testuser12",
8     "email": "test12@example.com",
9     "first_name": "Test",
10    "last_name": "User",
11    "role": "student"
12  }
13 }
```

Runner Capture requests Desktop Agent Cookies Vault Trash

If it didn't work (409 Conflict):

```
{  
  "error": "Username or email already exists"  
}
```

How to Log In

POST `/api/users/login` 

What it does: Get your login token

Do you need to be logged in? No

What to send: `application/json`

What to send:

```
{  
  "username": "student1",  
  "password": "password"  
}
```

Note: You can use your username or email address

Log In - What You Get Back

If it worked (200 OK):

```
{
  "success": true,
  "message": "Login successful",
  "token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9...",
  "user": {
    "id": 3,
    "username": "student1",
    "email": "student1@ase230.edu",
    "first_name": "Jane",
    "last_name": "Doe",
    "role": "student"
  }
}
```


If it didn't work (401 Unauthorized):

```
{  
  "error": "Invalid credentials"  
}
```

How to See Your Profile

GET `/api/users/profile` 

What it does: Show your info and your classes

Do you need to be logged in? Yes

How: GET

What to send:

```
Authorization: Bearer <your_token>
```

What you get: Your profile + your classes (if you're a student) or classes you teach (if you're a teacher)

Your Profile - What You Get Back

If it worked (200 OK):

```
{
  "success": true,
  "user": {
    "id": 3,
    "username": "student1",
    "email": "student1@ase230.edu",
    "first_name": "Jane",
    "last_name": "Doe",
    "role": "student",
    "created_at": "2025-10-15 10:30:00"
  },
  "enrollments": [
    {
      "id": 1,
      "title": "Web Development Fundamentals",
      "course_code": "ASE230",
      "credits": 3,
      "semester": "Fall",
      "year": "2025",
      "enrolled_at": "2025-10-15 10:30:00",
      "status": "active"
    }
  ]
}
```

← → ↺

bigbrain9842-796981.postman.co/workspace/83f4eff3-7005-4483-8c90-bc077f57cb29/request/49376604-e3190f57-35f5-4191-9962-7c639892319b?tab=headers

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🕒 History

🧩

GET Get data +

My Collection / Get data

Save ▾ Share 🔗

GET ▾ http://localhost:8080/api/users/profile Send ▾

Params Authorization Headers (11) Body • Scripts • Settings

Headers 🔍 10 hidden

	Key	Value	Description	⋮ Bulk Edit Presets ▾
☑	Authorization	Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJ1c2VyX2lkIjo2LCJ1c2Vybm...		
	Key	Value	Description	

Body Cookies (1) Headers (15) Test Results (1/1) | 🔄

200 OK • 92 ms • 778 B • 🌐 📄 Save Response ⋮

{} JSON ▾ ▶ Preview 🖼 Visualize ▾

```
1 {
2   "success": true,
3   "user": {
4     "id": 6,
5     "username": "testuser12",
6     "email": "test12@example.com",
7     "first_name": "Test",
8     "last_name": "User",
9     "role": "student",
10    "created_at": "2025-10-18 12:37:48"
11  },
12  "enrollments": [],
13  "courses_taught": []
14 }
```

🏠 🟢 Online 📄 Console

🏃 Runner 🐞 Capture requests 🖱 Desktop Agent 🍪 Cookies 🗑 Vault 🗑 Trash ?

How to Create a Class

POST `/api/courses` 

What it does: Make a new class (teachers and admins only)

Do you need to be logged in? Yes

What to send: `application/json`

What to send:

```
{
  "title": "Advanced Web Development",
  "course_code": "ASE400",
  "description": "Advanced concepts in web development",
  "credits": 4,
  "semester": "Spring",
  "year": 2026
}
```


Create Class - What You Get Back

If it worked (201 Created):

```
{
  "success": true,
  "message": "Course created successfully",
  "course": {
    "id": 4,
    "title": "Advanced Web Development",
    "description": "Advanced concepts in web development",
    "instructor_id": 2,
    "course_code": "ASE400",
    "credits": 4,
    "semester": "Spring",
    "year": "2026",
    "created_at": "2025-10-15 11:00:00",
    "updated_at": "2025-10-15 11:00:00",
    "first_name": "John",
    "last_name": "Smith",
    "username": "instructor1"
```

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POST Get data

My Collection / Get data

POST http://localhost:8080/api/courses

Send

Params Authorization Headers (12) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "title": "Test Course",
3   "course_code": "ASE919",
4   "description": "Test description"
5 }
```

Body Cookies (1) Headers (15) Test Results (0/1)

201 Created • 88 ms • 917 B • Save Response

{ } JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Course created successfully",
4   "course": {
5     "id": 5,
6     "title": "Test Course",
7     "description": "Test description",
8     "instructor_id": 2,
9     "course_code": "ASE919",
10    "credits": 3,
11    "semester": null,
12    "year": "2025",
13    "created_at": "2025-10-18 12:50:19",
14    "updated_at": "2025-10-18 12:50:19",
15    "first_name": "John",
16    "last_name": "Smith",
17    "username": "instructor1"
18  }
19 }
```

Online Console

Runner Capture requests Desktop Agent Cookies Vault Trash

If it didn't work (403 Forbidden):

```
{  
  "error": "Insufficient permissions"  
}
```

How to See All Classes

GET /api/courses

What it does: Show all available classes

Do you need to be logged in? No

How: GET

You can filter by:

- **semester** - Show only certain semester
- **year** - Show only certain year
- **instructor** - Show only certain teacher

Example:

```
GET /api/courses?semester=Fall&year=2025
```

See All Classes - What You Get Back

If it worked (200 OK):

```
{
  "success": true,
  "courses": [
    {
      "id": 1,
      "title": "Web Development Fundamentals",
      "description": "Introduction to web development",
      "instructor_id": 2,
      "course_code": "ASE230",
      "credits": 3,
      "semester": "Fall",
      "year": "2025",
      "created_at": "2025-10-15 10:00:00",
      "updated_at": "2025-10-15 10:00:00",
      "first_name": "John",
      "last_name": "Smith",
      "instructor_username": "instructor1",
      "enrollment_count": 2
    }
  ]
}
```

← → ↺

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GET Get data

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🔗 Share

🔗

My Collection / Get data

GET ▾ http://localhost:8080/api/courses

Send ▾

Params

Authorization

Headers (12)

Body ●

Scripts ●

Settings

Auth Type

Inherit auth from parent ▾

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).

No Auth

This request does not use any authorization. ⓘ

Body

Cookies (1)

Headers (15)

Test Results (1/1)

🔄

200 OK • 5 ms • 1.92 KB • 🌐 | 📄 Save Response •••

{} JSON ▾

▶ Preview

🖼 Visualize ▾

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How to Change a Class

PUT `/api/courses/{id}` 

What it does: Update an existing class

Do you need to be logged in? Yes

What to send: `application/json`

What to send:

```
{  
  "title": "Updated Web Development Course",  
  "description": "Updated course description",  
  "credits": 4  
}
```

Who can do this: Only the teacher who made the class or admins

Change Class - What You Get Back

If it worked (200 OK):

```
{
  "success": true,
  "message": "Course updated successfully",
  "course": {
    "id": 1,
    "title": "Updated Web Development Course",
    "description": "Updated course description",
    "instructor_id": 2,
    "course_code": "ASE230",
    "credits": 4,
    "semester": "Fall",
    "year": "2025",
    "created_at": "2025-10-15 10:00:00",
    "updated_at": "2025-10-15 12:00:00",
    "first_name": "John",
    "last_name": "Smith",
    "username": "instructor1"
```

← → ↺

bigbrain9842-796981.postman.co/workspace/83f4eff3-7005-4483-8c90-bc077f57cb29/request/49376604-e3190f57-35f5-4191-9962-7c639892319b?tab=body

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🔍 Search collections

📁 My Collection

GET Get data

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📁

PUT Get data

+

📄 Save ▾

Share 🔗

My Collection / Get data

PUT ▾

http://localhost:8080/api/courses/1

Send ▾

Params

Authorization

Headers (12)

Body

Scripts

Settings

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

☐ GraphQL

JSON ▾

Schema

Beautify

1 {

2 "title": "Updated Course Title",

3 "description": "Updated description"

4 }

Body

Cookies (1)

Headers (15)

Test Results (1/1)

🔄

200 OK • 135 ms • 926 B • 🌐 | 📄 Save Response ...

{} JSON ▾

▶ Preview

🖼 Visualize ▾

1 {

2 "success": true,

3 "message": "Course updated successfully",

4 "course": {

5 "id": 1,

6 "title": "Updated Course Title",

7 "description": "Updated description",

8 "instructor_id": 2,

9 "course_code": "ASE230",

10 "credits": 3,

11 "semester": "Fall",

12 "year": "2025",

13 "created_at": "2025-10-18 11:45:20",

14 "updated_at": "2025-10-18 12:45:41",

15 "first_name": "John",

16 "last_name": "Smith",

17 "username": "instructor1"

📄 Online

📄 Console

🏃 Runner

📄 Capture requests

🖱 Desktop Agent

🍪 Cookies

🔒 Vault

🗑 Trash

🔍 ?

If it didn't work (404 Not Found):

```
{  
  "error": "Course not found"  
}
```

How to Join a Class

POST `/api/enrollments` 

What it does: Sign up for a class

Do you need to be logged in? Yes

What to send: `application/json`

What to send:

```
{  
  "course_id": 2  
}
```

Note: You can only join classes for yourself (students only)

Join Class - What You Get Back

If it worked (201 Created):

```
{
  "success": true,
  "message": "Successfully enrolled in course",
  "enrollment": {
    "id": 5,
    "student_id": 3,
    "course_id": 2,
    "enrolled_at": "2025-10-15 12:30:00",
    "status": "active",
    "title": "Database Design",
    "course_code": "ASE240",
    "credits": 3,
    "semester": "Fall",
    "year": "2025",
    "first_name": "John",
    "instructor_name": "Smith"
```

← → ↺

bigbrain9842-796981.postman.co/workspace/83f4eff3-7005-4483-8c90-bc077f57cb29/request/49376604-e3190f57-35f5-4191-9962-7c639892319b?tab=body

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📁My Collection

- 🟢GET Get data
- 🟠POST Post data

📁Environments

- 📄Flows
- 🕒History
- 📁

POSTGet data⬵+

🔗My Collection / Get data

📁 Save ▾Share🔗

POST ▾http://localhost:8080/api/enrollments

Send ▾

ParamsAuthorizationHeaders (12)Body •Scripts •Settings

☐ none☐ form-data☐ x-www-form-urlencoded☒ raw☐ binary☐ GraphQLJSON ▾

1
2
3

{
 "course_id": 2
}

BodyCookies (1)Headers (15)Test Results (0/1)🔄

201 Created • 31 ms • 874 B • 🌐📄 Save Response ⋮

{}JSON ▾▶️ Preview🖼️ Visualize ▾

1 {
2 "success": true,
3 "message": "Successfully enrolled in course",
4 "enrollment": {
5 "id": 5,
6 "student_id": 6,
7 "course_id": 2,
8 "enrolled_at": "2025-10-18 12:50:48",
9 "status": "active",
10 "title": "Database Design",
11 "course_code": "ASE240",
12 "credits": 3,
13 "semester": "Fall",
14 "year": "2025",
15 "first_name": "John",
16 "instructor_name": "Smith"
17 }
}

📄🟢 Online📄 Console

🏃 Runner🐞 Capture requests🖱️ Desktop Agent🍪 Cookies🔒 Vault🗑️ Trash🔧?

30

If it didn't work (409 Conflict):

```
{  
  "error": "Already enrolled in this course"  
}
```

How to See Your Classes

GET `/api/enrollments` 

What it does: Show all the classes you joined

Do you need to be logged in? Yes

How: GET

You can filter by:

- `student_id` - Show classes for a certain student (admins and teachers only)
- `course_id` - Show only a certain class
- `status` - Show only active or inactive classes

See Your Classes - What You Get Back

If it worked (200 OK):

```
{
  "success": true,
  "enrollments": [
    {
      "id": 1,
      "student_id": 3,
      "course_id": 1,
      "enrolled_at": "2025-10-15 10:30:00",
      "status": "active",
      "title": "Web Development Fundamentals",
      "course_code": "ASE230",
      "credits": 3,
      "semester": "Fall",
      "year": "2025",
      "instructor_first_name": "John",
      "instructor_last_name": "Smith"
    }
  ]
}
```

← → ↺

bigbrain9842-796981.postman.co/workspace/83f4eff3-7005-4483-8c90-bc077f57cb29/request/49376604-e3190f57-35f5-4191-9962-7c639892319b?tab=headers

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📁 Environments

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GETGet data

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🔗 My Collection / Get data

📄 Save⌵

Share🔗

GET⌵

http://localhost:8080/api/enrollments

Send⌵

Params

Authorization

Headers (12)

Body ●

Scripts ●

Settings

Cookies

Headers

👁 10 hidden

	Key	Value	Description	⋮ Bulk Edit Presets ⌵
☑	Authorization	Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJ1c2VyX2lkIjo2LCJ1c2Vybm...		
⋮	Content-Type	application/json		🔗 Type 🗑️
	Key	Value	Description	

Body

Cookies (1)

Headers (15)

Test Results (1/1)

🔄

200 OK • 7 ms • 854 B • 🌐 | 📄 Save Response ⋮

{} JSON ⌵

▶ Preview

🖼 Visualize ⌵

```
1 {
2   "success": true,
3   "enrollments": [
4     {
5       "id": 5,
6       "student_id": 6,
7       "course_id": 2,
8       "enrolled_at": "2025-10-18 12:50:48",
9       "status": "active",
10      "title": "Database Design",
11      "course_code": "ASE240",
12      "credits": 3,
13      "semester": "Fall",
14      "year": "2025",
15      "instructor_first_name": "John",
16      "instructor_last_name": "Smith"
17    }
18  ]
19 }
```

📄

📄 Online

📄 Console

🏃 Runner

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📄 Desktop Agent

📄 Cookies

📄 Vault

🗑️ Trash

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When Things Go Wrong

Common Error Codes

- **200 OK** - Everything worked fine
- **201 Created** - Something new was made successfully
- **400 Bad Request** - You sent bad information
- **401 Unauthorized** - You need to log in first
- **403 Forbidden** - You don't have permission
- **404 Not Found** - We can't find what you're looking for
- **409 Conflict** - This already exists
- **500 Internal Server Error** - Something broke on our end

Error Messages

What error messages look like:

```
{  
  "error": "Error message description"  
}
```

Examples:

```
{  
  "error": "Missing required fields: username, password"  
}
```

```
{  
  "error": "Authentication required"  
}
```

Testing the Website

How to test with cURL

Sign Up:

```
curl -X POST http://localhost:8080/api/users/register \
  -H "Content-Type: application/json" \
  -d '{"username": "testuser", "email": "test@example.com", "password": "password123", "first_name": "Test", "last_name": "User"}'
```

Get your profile (you need to be logged in):

```
curl -X GET http://localhost:8080/api/users/profile \
  -H "Authorization: Bearer <your_token>"
```

Testing Tools

What you can use to test

1. cURL Scripts - Located in `/tests/curl/`

- Test each part one by one
- Test everything at once (`test_all_apis.sh`)

2. Web Test Page - `/tests/html/api_test_suite.html`

- Test using a web page
- See results right away

3. Website Help - `http://localhost:8080/api/`

- See all the things you can do
- See examples of requests and responses

Security Features

How we keep things safe

- **Login Tokens** - Safe password system
- **Different User Types** - Student, Teacher, Admin roles
- **Permission Checks** - Only certain people can do certain things
- **Password Protection** - Passwords are stored safely
- **Input Checking** - We check all the information you send
- **Cross-Origin Support** - Works from different websites

Database Tables

What information we store

- **users** - User accounts and login info
- **courses** - Class information
- **enrollments** - Which students are in which classes
- **assignments** - Class homework
- **grades** - Student grades
- **quizzes** - Quiz information
- **quiz_questions** - Quiz questions
- **quiz_answers** - Quiz answer choices
- **notifications** - Messages for users

Best Ways to Use This

Tips for using the website

1. **Always include Content-Type header** when sending information
2. **Keep your login token safe** and put it in the Authorization header
3. **Check if things worked** - look at the response codes
4. **Check your information** before sending requests
5. **Use HTTPS in real life** for safe communication
6. **Handle errors nicely** in your applications

Summary

What we built

- ✓ 8 Complete Website Features
- ✓ Safe Login System
- ✓ Different User Types
- ✓ Good Error Handling
- ✓ Full Create/Read/Update/Delete
- ✓ Lots of Testing Tools

What's next

- Make a website frontend
- Add more features (homework, grades)

Thank You!

Questions & Help

Code Repository: <https://github.com/BsalBhandari/ase230-bhandari-project1>

Website Help: `http://localhost:8080/api/`

Test Tools: `/tests/` folder

Extra: Complete Test Commands

Running All Tests

```
# Make the test file work
chmod +x tests/curl/test_all_apis.sh

# Run all tests
./tests/curl/test_all_apis.sh
```

What should happen: All 8 website features should work  and show proper JSON responses with login working correctly.